

Vincent Zhou

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EDUCATION

University of Waterloo
Bachelor of Applied Science, Computer Engineering, Honours Co-op

Sept 2025 – June 2030

EXPERIENCE

Waterloo Aerial Robotics Group (WARG)

Autonomy Project Manager

Sept 2026 – Present

- Lead a **30-member** autonomy subteam, organizing contributors into parallel project groups, scoping the flight-software backlog into issues sized for onboarding, and tracking them through to flight readiness for the 2027 AEAC competition.
- Run code review for the autonomy stack and set its architectural direction, defining the ROS 2 node and message boundaries between the perception, GNC, and obstacle-avoidance packages so work split across groups integrates without rework.

Software Engineering Intern

May 2026 – Sept 2026

- Independently built and flight-tested a **visual teach-and-repeat** precision-landing service 🌀 in **ROS 2** as concurrent action servers (Takeoff/Landing) over a multi-threaded executor with reentrant and mutually-exclusive callback groups, fusing camera, IMU, GPS, and rangefinder streams into a **10 Hz** control loop with cancellation support for pilot override.
- Designed the underlying algorithm: an altitude-keyed map of **ORB** features recorded during takeoff, matched on descent against the nearest stored frame via Lowe-ratio-filtered kNN, back-projected to metric ground coordinates, and resolved with **RANSAC** affine estimation, returning the drone to its launch point within **15 cm in flight test** and **2 cm in simulation**.
- Deployed to an **NVIDIA Jetson**: containerized the **CUDA**-accelerated **OpenCV** build, wrote the Arducam V4L2 driver plus systemd service and **Bash** provisioning and health-check scripts, and diagnosed a native library-ordering fault that crashed the camera pipeline at initialization.
- Engineered the failure path with a stale-vision recovery state machine, inlier-ratio and scale validation to reject degenerate matches, and a debounced handoff to **ArduPilot**'s LAND mode so the autopilot owns touchdown.
- Architected a full-stack, real-time **UAV geolocation platform** (**React/TypeScript** frontend, **Flask** REST API) for the 2026 AEAC Firefighting Competition, enabling ground operators to capture, annotate, and geolocate targets spotted mid-flight using a pinhole camera model and ray-plane intersection against live telemetry to localize targets in 3D to **<0.5 m**.
- Designed the REST surface (capture, annotate, generate-output, history, export) and a custom drag-to-pan / scroll-to-zoom annotation canvas with live telemetry readouts, giving operators a human-in-the-loop review pipeline.

PROJECTS

Classly 🌀 | *Classroom Management Platform: React 19, Node.js/Express 5, Postgres*

- Built a full-stack school management platform with a **Postgres/Drizzle ORM** schema, role-based access control (admin/teacher/student), and **Better-Auth** session authentication (email/password, Google/GitHub OAuth).
- Migrated the backend off a long-running **Render** container onto **Cloudflare Workers**, moving to serverless Postgres over WebSockets, scoping connection-pool lifetime to the request to fix pools leaking across transactions, adding a cron-triggered session-cleanup job, and managing secrets through **Wrangler**.
- Hardened the API with **Arcjet** bot detection and rate limiting, parameterized queries, and row-level locking with transactional rollback to prevent race conditions during concurrent enrollment.

SITL-Plus 🌀 | *UAV Software-in-the-Loop Simulator, WARG*

- Built a from-scratch simulator using **PyBullet** rigid-body physics to model quadcopter flight dynamics, bridged to **ArduPilot**'s SITL binary over a **UDP** JSON protocol for high-rate physics/flight-controller state exchange.
- Simulated onboard sensors: downward and forward RGB-D cameras via the OpenGL renderer and a ray-cast rangefinder/LiDAR, streamed to **Rerun** over **gRPC** for live 3D visualization, enabling integration testing of the vision and landing pipelines without hardware.
- Containerized the stack (physics engine + ArduPilot SITL/MAVProxy) with **Docker Compose**, covered the sensor and dynamics models with a **pytest** suite, and used the simulator to validate and tune the precision-landing pipeline to **2 cm** before committing to flight tests.

CNN Feature Matching for Precision Landing | *extends the WARG landing pipeline*

Sept 2026 – Present

- Refactoring the landing pipeline to run feature extraction behind a swappable backend interface, then benchmarking a learned 2D CNN detector-descriptor (**XFeat**) against hand-engineered **ORB** on match quality under viewpoint and illumination change and on inference latency against the on-board **10 Hz** budget.

Health Pro Wellness Clinic 🌀 | *Astro, Cloudflare Pages*

- Shipped a production trilingual (EN / Simplified / Traditional Chinese) static site for a local clinic on **Cloudflare Pages**, with a CI audit harness enforcing WCAG AA contrast, a hash-based **CSP**, transfer-weight budgets, and layout regression across 11 viewports via headless Chrome.

SKILLS

- **Languages:** Python, C++, TypeScript/JavaScript, Java, SQL, Bash, Lua, Verilog
- **Robotics & Vision:** ROS 2, ArduPilot/MAVROS, MAVLink, OpenCV, CUDA, PyBullet, Rerun, NumPy, Eigen, NVIDIA Jetson, Raspberry Pi
- **Backend & Data:** FastAPI, Flask, Node.js/Express, REST, PostgreSQL, Drizzle ORM, Neon, Pydantic, Zod
- **Infrastructure & Testing:** Docker, Cloudflare Workers/Pages, Wrangler, Git/GitHub, GitHub Actions CI, Linux/systemd, pytest, Vitest, Quartus/FPGA